

Advanced SOC Practical Applications & Incident Response

1. Security Tools Advanced Practicals

- **Snort Rule Testing:** Install Snort and create detection rules.

Example Rule:

```
alert tcp any any -> any 80 (msg:"Malicious Domain"; content:"malicious.com"; http_uri; sid:1000001;)
```

Test with: curl http://malicious.com

- **Nessus Scan:** Run Nessus Essentials against Metasploitable2. Export vulnerabilities report.

Top 3 vulnerabilities by CVSS Score (example):

Vulnerability	CVSS Score
vsftpd Backdoor (CVE-2011-2523)	10.0
Unpatched Apache Server	9.8
Weak SSH Ciphers	7.5

- **Osquery Monitoring:** Install Osquery on Windows VM. Query running processes:

SQL Example: SELECT * FROM processes;

Simulate malicious process with harmless batch script.

2. Security Operations Workflow Simulation

Workflow stages simulated with TheHive:

1. Detection → Alert received (failed logins + malicious URL)
2. Triage → Analyst assigns priority
3. Investigation → Check logs, correlate IOCs
4. Response → Block IP, terminate malicious process
5. Documentation → Case added to TheHive

3. Situation Report (SITREP)

Timestamp	Event	Action Taken
2025-10-05 14:10	Multiple failed logins from 192.168.1.50	Blocked IP at firewall
2025-10-05 14:25	User clicked phishing link	Reset account credentials
2025-10-05 14:40	Suspicious process detected	Terminated process via EDR

4. Incident Response Playbook (Phishing)

Pseudocode workflow for phishing attack:

IF suspicious email detected: - Quarantine email - Alert SOC Tier-1 Analyst - Analyst reviews headers and links
IF confirmed phishing: - Block sender domain - Force password reset for user - Create incident in TheHive - Notify stakeholders

5. Threat Hunting Exercise

Hypothesis: Unauthorized privilege escalation attempts.

Search Elastic SIEM for Event ID 4672 (special privileges assigned).

Query Example: index=security-* EventID=4672 | stats count by User

6. Velociraptor Netstat Sample

Proto	Local Address	Foreign Address	State
TCP	0.0.0.0:80	192.168.1.10:52344	ESTABLISHED
TCP	0.0.0.0:445	192.168.1.15:60012	TIME_WAIT

7. Memory Forensics (Placeholder)

SHA256 hash of memory dump (example placeholder):

3d2f4e7a8c9b1d2233445566778899aabbccddeeff112233445566778899aabb